

FES: A modular pipeline to find, encode and select relevant facts from clinical notes

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Abstract—x

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I. INTRODUCTION

Epilepsy is one of the most common neurological diseases, affecting over [x]m people globally. Clinicians use the type of epilepsy, type of seizures and frequency of seizures to determine the severity of the patient’s disease. [1] Clinical trials typically use seizure freedom as their target outcome. [2] The only place this information is routinely recorded is in clinic letters from an epilepsy specialist to the patient’s primary care doctor after the patient visits the clinic. To gather this information at scale, experts are required to complete manual chart review: review each letter, interpret the key details and translate them into a standard form. This process is costly, time-consuming, and error prone. [3] Making this information available in a structured form that you can query in a database to summarise, group and filter can make large-scale research efforts viable.

To make this type of structured data available, there are two broad approaches: collect the data in a structured form or process unstructured data into that form. It is simple to collect medication in a more structured form by using separate fields for medication, dose, unit and frequency, rather than writing ‘Topiramate 100 mg BD’. Applying that to seizure frequency is more difficult because they are not simple facts to record in a single, consistent form. A typical example is, *“focal impaired-awareness seizures have been occurring more often, 2 times per week, whereas previously they only happened once or twice a month.”* Imprecise language like this is known as hedging[5], and paradoxically, stripping it down to a single number would make it less precise and less useful. A system that enables clinicians to record these richer observations and then processes them into a standardised form is ideal.

Natural language processing (NLP) and the subdomain of information extraction (IE) offers a solution for this. Rules-based methods have successfully extracted information on medications and diagnoses, but struggled with the imprecise language and diverse forms of seizure frequency statements. ‘2 to 3 seizures a week’, ‘frequent bouts of seizures’ and ‘4 since her last visit’ are all valid descriptions of the same fact. Large language models (LLMs) have demonstrated significant improvements in semantic understanding than prior methods and have successfully extracted seizure frequency information, but they occasionally hallucinate facts. Combining the semantic

understanding of an LLM with the transparency of a rules-based system could provide the most accurate, trustworthy results.

Effective IE can be broken down into three key stages: finding relevant candidate facts, encoding those facts into a particular form, and selecting which facts to keep and which to filter out. A clinic letter can contain multiple descriptions of seizures, some are historical events while others are current, some are certain seizures while others may be “absence-like episodes”, some are describing the same event but in another context. Seizure frequencies can be expressed as a rate, a vague quantity and a duration. A transparent system would allow you to see not just the final answer, but which steps were taken to achieve it.

This paper evaluates the ideal combination of LLMs and rules across those three stages: find, encode and select. It was evaluated on a set of 1,200 annotated epilepsy clinic letters. The paper demonstrates that a modular system combining LLMs and rules can extract information with comparable accuracy to past methods while preserving a richer evidence chain. The system is available as an open source package at [INSERT LINK](#). Figure 1 displays the web app illustrating some of that chain from evidence to final answer.

II. LITERATURE REVIEW

NEED A BETTER SECTION ON TASK DECOMPOSITION

Wang *et al.* define clinical IE as the automatic extraction and encoding of clinical information from text [8]. Fu *et al.* separate mention detection from concept encoding: the quoted span is distinct from the coded concept [6]. Xie *et al.* extract a frequency sentence and then apply rules to encode it into a number [9]. In much of the remaining epilepsy literature encoding is bundled into the extraction phase, with the model extracting e.g. ‘4 per month’ from a raw statement ‘about once a week’. When rules are used to format the answers, they are often described as simple post-processing steps for scoring instead of distinct stages in the pipeline. Those paper shows that separating find and encode as separate stages preserves more of the evidence while still being able to encode it into the required form for downstream use.

DECIDE WHAT TO SAY ABOUT IAA

A number of methods have shown performance at or above human level. Fonferko-Shadrach *et al.* reported the annotators found annotation difficult and seizure frequency was the most difficult. This was reflected in the inter annotator agreement

(IAA) score of 0.47 (moderate agreement) even after multiple phases of development of the guidelines to meet the annotators’ needs. Xie *et al.* reported an overall pairwise F1 score of 0.44 on a task focused extracting seizure freedom, frequency and most recent seizure statements and recommend double or triple annotation. This is a long-standing challenge in the field, as Grisham and Sundheim noted in 1996 ‘it proved remarkably difficult to formulate guidelines which were reasonably complete and consistent’. Despite levels of IAA of 0.5, the literature commonly reports that rules-based and LLM-based methods achieving F1 scores of 0.8 and above.

INSERT A REVIEW OF EVALUATION PERFORMANCE

We have 8 different papers in the last 4 years with seizure frequency performance results, but there is significant variance.

FINE-TUNING VS. IN-CONTEXT LEARNING / PROMPTING

Chiticariu *et al.* show that academic papers tend to focus exclusively on benchmark accuracy, while practitioners tend to value transparency, configurability and efficiency at the cost of some accuracy. Fu *et al.* show that this translates to the clinical setting too, and model interpretability is also a critical factor in successful deployment in clinical settings. Clinicians are highly trained in using deductive logic to draw conclusions and they expect systems to expose transparent, interpretable evidence that they can reliably use to make decisions that demand high accountability. Fonferko-Shadrach *et al.* expose evidence spans as a consequence of using a rules-based system, and Gan *et al.* utilise evidence spans as a training signal. This paper presents a transparent evidence chain from exact text to final answer as part of the core output to improve interpretability for downstream users.

III. METHODS

A. Data

The dataset is a collection of epilepsy clinic letters which record a UK patient’s consultation with an epilepsy specialist, often a neurologist or specialist nurse. [2], [3] The NHS requires that clinic letters are sent to the patient’s GP after all outpatient visits such as these. [13] They are often the only record of a patient’s diagnoses, treatments and outcomes in neurological disorders, and they are written in narrative form. An example clinic letter is shown in Figure 1.

Gan *et al.* produced this dataset of synthetic letters modelled on real clinic letters from King’s College London hospital. The real letters were used to create 10 template letters and they were combined with 500 short descriptions of seizure frequency statements produced by GPT-5 to create the full set of 15,099 letters. A sample of 1,200 letters was used in this paper. Each seizure frequency statement was annotated by trained experts to create the gold-standard labels. Each letter has a single label representing the patient’s current seizure frequency burden.

Minimal preprocessing was applied to the raw data. Letters and annotations were stored both in a single .json file. Letters are minimally cleaned to correctly parse quotes, new lines,

Thank you for asking me to review your patient. I am concerned that her epilepsy has worsened in recent months, with a clear pattern of deterioration. She and her partner report increasing clusters of events correlating with the perimenstrual period, suggesting a catamenial component. In particular, she describes brief focal episodes with preserved awareness that tend to occur as she is settling to sleep, at the point of drifting off, now happening on most nights of the week. These events are accompanied by a rising epigastric sensation, facial pallor, and right-hand automatisms lasting 30–60 seconds, followed by fatigue and headache. In addition, she has had three generalised tonic–clonic seizures in the last six weeks, two of which occurred during the same perimenstrual window.

Fig. 1: An example clinic letter with three distinct seizure frequency references highlighted as potential candidates.

TABLE I: Dataset summary

	Development	Test
Number of letters	750	450
% of letters with	a frequency (count, rate)	62%
	seizure free period	15%
	unknown seizure frequency	13%
	no seizure frequency	10%

tabs, the $\leq$ sign and nulls. The raw letters were ingested in the pipeline in full; no tokenisation, section-splitting or truncation was applied.

The dataset was split up into stratified samples for separate development and test sets. It was divided into 750 development letters and 450 test letters. They were stratified by the kind of seizure frequency fact so both sets consist of 62% with a frequency, 15% seizure free, 13% unknown and 10% with no seizure frequency reference. Table I summarises those details.

B. Architecture

To trace the journey from raw evidence to final answer, the task was divided into three key stages: find, encode, select. First, the pipeline identifies potential candidates that fit the description in the raw text. Next, the pipeline takes the raw text from those candidates and encodes them into the required form. Finally, the pipeline determines which of those encoded facts meet the criteria for the final output. The specific configuration of the each stage is determined by the nature of the task: which labels to extract from the text, what schema to encode them into, and how many labels to output at the final stage. The same modular architecture effectively describes multi-label tasks such as inventory and single label tasks such as current frequency. Figure 2 illustrates the general architecture of the pipeline.

There are two key benefits of breaking the task into three discrete stages: configurability and transparency. Past research has shown that entirely rules-based methods can successfully populate the inventory, while entirely LLM-based methods can successfully extract the current frequency. While some

TABLE II: Pipeline method

Method	Find	Encode	Select
Rules only	Rules	Rules	Rules
LLM + Rules	LLM	LLM + Rules	Rules
LLM only	LLM*		LLM

*A single LLM prompt is used for the find and encode stages. A second LLM prompt is used for select.

methods have used a combination of rules and LLMs to produce the final answer, none have empirically assessed the ideal combination of the two. This three-stage pipeline enables evaluation of the best combination overall, and specifically which component performs best at which stage. Table II summarises the 3 main methods evaluated in this paper.

Determining which one fact among many is correct is the challenge. Sometimes that requires grouping together multiple related facts into a single answer, e.g. '1 in June, 0 in July, 2 in August' becomes '3 in 3 months'. Sometimes it requires determining which fact takes precedence when there are multiple competing facts with varying degrees of specificity and certainty e.g. 'She may remain seizure-free for up to 4 month, but then will experience clusters of 5 seizures in a single day'. Different annotation guidelines would record ambiguous facts like these differently. Distinct stages in the pipeline make those choices configurable and explicit.

NEEDS A COMPLETE PORTRAYAL OF EACH STAGE?

The development set was used to optimise the two key elements of the pipeline: the LLM prompt instructions and the rules. The focus of the optimisation process was to improve accuracy while . Qualitative error analysis was conducted to diagnose the cause of poor performance in particular areas and identify common error modes. Hypotheses were developed and tested to tune the prompt and rules iteratively.

Through iterative error analysis on both the LLM and the rules, it became apparent that the complexity of the task was often a function of competing rules and guidelines. Often, improving a rule or a prompt instruction to address one class of error would harm another area because annotation policies are complex. This complexity is a challenge for human annotators also; Fonferko-Shadrach *et al.* described seizure frequency as the most difficult category to annotate and their inter-annotator agreement of 0.47 shows annotators often disagreed about how to represent seizure frequency statements.

Decomposing the task into three stages allowed for cleaner separation of the policies on which types of facts to extract and which criteria to apply to exclude specific variants of them. Decomposing the task into three stages also made it possible to evaluate the performance at each individual stage and identify the bottlenecks in the system. It is this combination of a cleaner task decomposition for the pipeline and a clearer view into the pipeline that enabled steady, incremental gains to materialise from the optimisation process.

C. Evaluation

To enable a direct comparison with the results in Gan *et al.*, the pipeline is evaluated using the same micro-F1 score.

TABLE III: Seizure frequency categories: Purist and Pragmatic

Purist	Pragmatic
Daily	Frequent
More than weekly, less than daily	
Once a week	
More than monthly, less than weekly	
Once a month	Infrequent
More than every 6 months, less than monthly	
Once every 6 months	
Less than once every 6 months	
Unknown	Unknown
No seizure frequency reference	No seizure

They calculate the F1 on two categories; a fine-grained 'purist' score and a coarser 'pragmatic' score. The more precise purist F1 is the primary metric in this paper, but the pragmatic F1 score is used for frequency bands with very small sample sizes (<20). A full specification of the categories is in Table III. The rationale for categorising labels is so that semantically identical phrases are matched when exact labels differ, so that daily, once per day and every day would be considered correct matches to the gold label '1 per day'. This evaluates the model on how effectively it can identify the correct frequency rather than replicate precise phrasing. Further explanation on the categories can be found in the original paper from Holgate *et al.*

The pipeline is evaluated on a held-out test set of 450 letters. These letters were never used to tune prompts, rules or any other aspect of the pipeline. The test set is therefore used to evaluate how effectively the model generalises to unseen data. These results are compared against the evaluation set labelled Synthetic (1,166) in Gan *et al.* which was also a held-out sample from the synthetic dataset, distinct from the training set used to fine-tune the models. It is important to note that Gan *et al.* also tested their results a distinct set of real clinic letters to demonstrate generalisation to real data; our results and comparisons are limited purely to the synthetic data.

The primary objective of this research is to identify which component performs best at which stage. Table II shows the 3 methods used in the evaluation: rules at every stage, the LLM at every stage or a combination of LLM and rules. Each method was evaluated at each stage: find, encode and select. This produces a 3x3 matrix of results that answer two key questions:

- 1) Among the three methods, which performs best: rules, LLM, or a combination of both?
- 2) How much incremental value is there from the additional encode and select stages?

The primary hypothesis is that a combination of an LLM plus rules will perform better than either on their own, because they have complementary strengths and those strengths are particularly well suited to distinct stages in the pipeline. The secondary hypothesis is that multiple discrete stages will improve performance for the LLM, regardless of whether the subsequent stages are rules or additional LLM calls. This

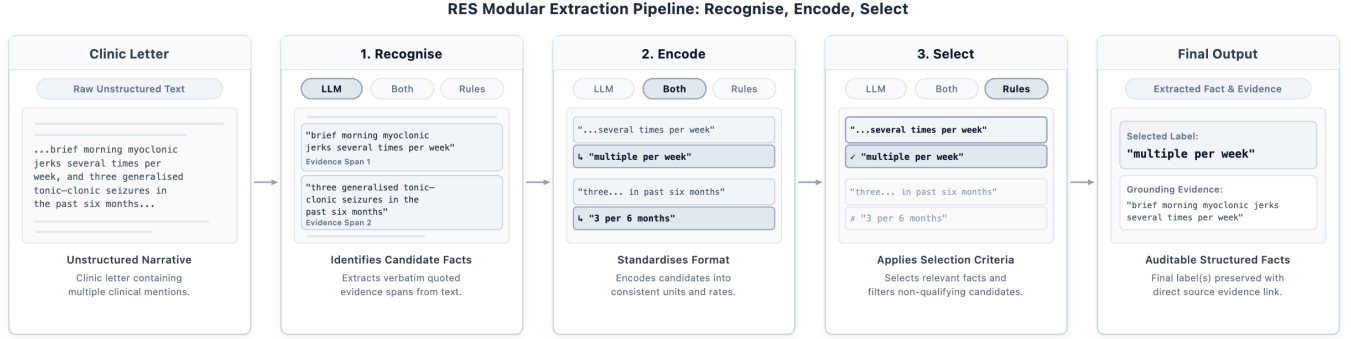


Fig. 2: A clinic letter passes through three stages in the pipeline - find, encode and select - to generate structured labels with the exact text used from the letter to produce the label. Each of the three stages can be configured to use an LLM, rules or both. The example shown is the task of extracting a single label representing the patient’s current seizure frequency burden.

builds on the idea that task decomposition is an effective form of context engineering, and is particularly useful in scenarios where the prompt is long and contains many layers of instructions.

IV. RESULTS

A. Development Environment

Experiments using local language models took place on a Dell XPS 16 9640 running Windows 11, with an Intel Core Ultra 9 185H, 32 GB RAM, and an NVIDIA GeForce RTX 4070 Laptop GPU (8 GB VRAM). Ollama v0.32.15 was used to serve the models. Two models were run on local hardware (Qwen 3.8 27b, Gemma 4 26B).

Experiments using hosted APIs were run from a Mac mini M1 with 16 GB unified memory on macOS 26.5.2. API calls were made directly to the provider (OpenAI, DeepSeek) or through model routers (OpenRouter, AI Gateway) Four hosted models were used (Gemini 3.7 Flash, GPT 5.6 Luna, Grok 4.6, DeepSeek v4 Flash).

All models used a temperature of 0 except Luna, which only accepts a temperature of 1. An ablation was run on Gemini using a temperature of 1. All models used a thinking or effort level of ‘low’. Ablations were run on Gemini using thinking levels of medium and high.

Python 3.14.4 was used to write the program. DSPy 3.2.1, Pydantic 2.13.4 and LiteLLM 1.87.1 were utilised as core packages. DSPy was used to co-ordinate the model calls and require structured JSON output. Pydantic was used to validate the structured output conforms to the specified schema. LiteLLM was used to deliver API calls to Ollama and the model routers.

B. Benchmark Performance

The LLM and rules method achieves state-of-the-art results (SOTA) for extracting seizure frequency from epilepsy letters. It achieved a purist micro-F1 score of 0.83, exceeding the prior benchmark performance of 0.81 in Gan *et al* set by a model that was fine-tuned and evaluated using this same

TABLE IV: Comparisons to past seizure frequency papers

Author	Year	Method	F1 score
Xie et al.	2022	Fine-tuned LM	0.83*
Decker et al.	2022	Rules	0.82**
Fonferko-Shadrach et al.	2024	Rules	0.66
Fernandes et al.	2024	Fine-tuned LM	0.37
Holgate et al.	2024	Prompted LLM	0.73
Holgate et al.	2025	Fine-tuned LLM	0.68
Gan et al.	2026	Fine-tuned LLM	0.81
Brown & Gan	2026	LLM and rules	0.83

*Drops to 0.73 on texts with a frequency statement.

**Drops to 0.4 on the external test set.

dataset of synthetic letters. While that is the only paper to evaluate the same dataset with the same metric, Table IV includes a summary of all 7 papers that reported a seizure frequency extraction score this decade.

This method is particularly effective at identifying high frequency seizure events. A common theme in previous papers is that their method performed best when the task was predicting ‘unknown’ or ‘no seizures’. Xie et. al reported an F1 score of 0.92 on texts that contained no seizure frequency statement and 0.73 when there was a seizure frequency reported. Gan et al. and Holgate et. al both report unknown as their highest scoring category at 0.89 and 0.87 respectively, while daily frequency (0.7, 0.41) and more than weekly (0.68, 0.55) are much lower. In this method, daily (0.92) and more than weekly (0.91) are the best performing categories, significantly above unknown (0.74). The full performance breakdown across purist and pragmatic categories can be seen in Table V.

C. Method Performance by Stage

The LLM and rules method performs better than the LLM or rules alone at every stage: find, encode, select (Table VI). The LLM is significantly better at the initial find stage vs. rules alone (0.79 vs 0.65). The LLM and rules combination then adds incremental gains at the encode (+0.01) and select (+0.03) stages, while the subsequent LLM stages make a very marginal difference (+0.006). This illustrates that the LLM

TABLE V: LLM and Rules: Purist and Pragmatic micro-F1

Purist	n	Purist F1	Pragmatic F1
Daily	41	0.92	Frequent: 0.95
More than weekly, less than daily	123	0.91	
Once a week	5	0.75	
More than monthly, less than weekly	58	0.9	
Once a month	18	0.73	Infrequent: 0.72
More than every 6 months, less than monthly	55	0.74	
Once every 6 months	1	0.00	
Less than once every 6 months	6	0.6	
Unknown	76	0.74	Unknown: 0.74
No seizure frequency reference	67	0.81	No seizure: 0.81

TABLE VI: Purist micro-F1 at find, encode, and select

Method	Find	Encode	Select
Rules only	0.65	0.65	0.72
LLM + rules	0.79	0.80	0.83
LLM only		0.79*	0.79

*A single LLM prompt is used for the find and encode stages.

and rules have complementary strengths that are particularly effective at discrete stages.

The LLM is particularly strong at the find stage: identifying high quality candidates in the text and providing an initial proposed answer in the required form. This initial stage is equivalent to the entire extraction pipeline in most prior papers: find, encode and select all at once. The LLM at this stage performs at a level comparable to the best prior methods, achieving performance in a single prompt that previously required fine-tuning.

Rules are particularly effective at the select stage: reviewing all of the candidates proposed by the model and determining which should be the final answer. This can mean either choosing a different candidate, such as choosing a frequency candidate when the model has chosen 'unknown', or performing a mathematical calculation on the selected evidence to update the answer, such as counting the total events over a monthly seizure diary. This is a direct consequent of the pipeline design. The find phase outputs multiple candidate events with evidence quoted directly from the text, rather than just a single flat answer. Rules leverage that richer evidence to deliver significant incremental gains.

D. Model Comparison

The LLM and rules method shows consistent, incremental performance gains on each of the six models evaluated (Figure 4). The average performance gain was +0.06 from find to select, with Luna showing the largest gain of +0.09. Model performance on purist F1 closely correlates to overall medical intelligence as measured by Artificial Analysis' Healthcare and Medical Index. They cluster into three groups: strong (Grok 0.84, Gemini 0.83), medium (DeepSeek 0.8, Luna 0.78), weak

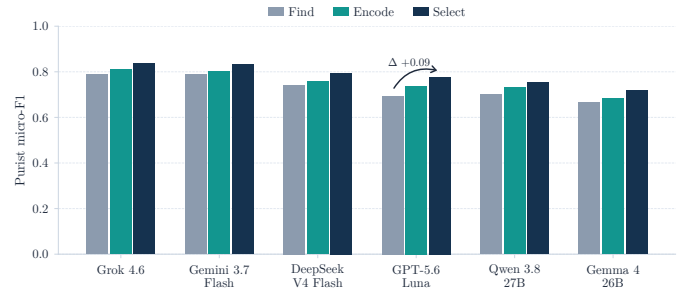


Fig. 3: LLM and rules. Every model sees consistent performance gains in the encode and select stages. Luna saw the biggest performance gain (0.69→0.78).

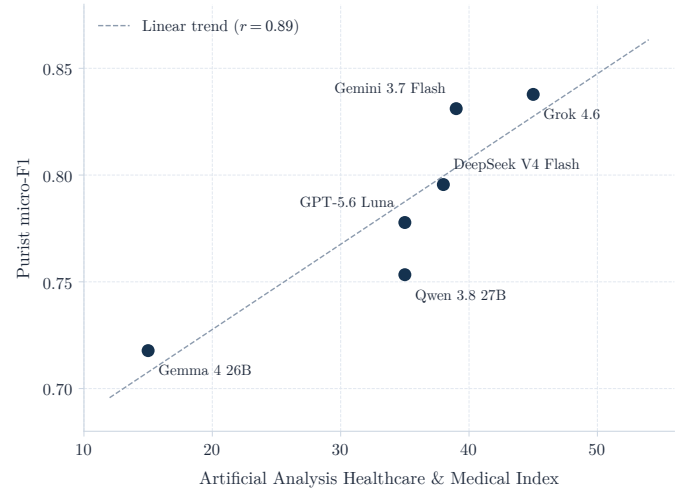


Fig. 4: The smartest models are the best frequency extractors.

(Qwen 0.75, Gemma 0.71) which largely reflect the model size and cost.

Local models that fit on consumer-grade hardware now deliver superior performance to models that required eight Nvidia V100 GPUs two years ago (Holgate, 2024). Qwen's performance of 0.75 was achieved running locally on a laptop with a single GPU, surpassing Llama 2's performance of 0.73. That is a critical development as it means hospitals can use their own servers to securely process patient data to reliably extract clinical information like this. DeepSeek is small enough for many hospital servers to host and it performs better than a model fine-tuned on 1,797 synthetic letters.

E. Model Configuration

Raising the reasoning effort level did not improve performance. Gemini's purist-F1 score was 0.83 at low and 0.82 at high; a paired difference of +0.01, 95% CI -0.01 to 0.04; $p = 0.39$. Extra reasoning adds cost and latency without improving extraction.

The temperature setting also had no significant effect on the results. The F1 score was 0.83 on a temperature of 0 versus 0.82 at a temperature of 1 (paired difference +0.01, 95% CI -0.02 to 0.03; $p = 0.71$). A temperature of 0 was an appropriate

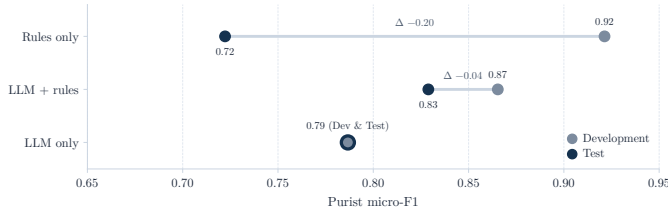


Fig. 5: Rules only performs best on the development set but overfit. LLM only generalises best with consistent performance. LLM and rules combined produce the best results overall, even with a generalisation gap of -0.04 (0.87 $\rightarrow$ 0.83).

default across models and Luna at a temperature of 1 is a valid comparison.

The task is tightly defined with explicit schema and output requirements, which limits the expected variability by design. This partly explains why neither setting had a significant effect on results and it indicates the robustness of the method.

F. Development vs. Test Results

Rules perform better on the development set but significantly worse on the test set (Figure 5), which repeats the pattern of overfitting commonly found in the literature. Rules are brittle and tend to overfit specific patterns in the training data, generalising poorly to unseen data. The rules only method achieves an F1 score of 0.92 on the development set, but performance declines significantly (-0.20) on the test set. LLM only achieves a consistent score of 0.79 across both splits which shows the method generalises well. The LLM and rules method combines the best parts of each: it performs significantly better on the development set than LLM only (+0.08) and it has a significantly smaller generalisation gap than rules only (-0.04). The LLM does the heavy lifting of matching patterns across the entire text, allowing the rules to focus on generalisable patterns in the refined evidence set.

G. Error Analysis

The method performed worst at extracting infrequent seizure events, mirroring the pattern in Gan et. al. The F1 score for the infrequent (monthly or less) category was 0.72. The main error pattern is in assigning unknown or no seizure frequency reference to letters with infrequent labels in the gold-standard annotations (Figure 6).

A few examples include:

V. DISCUSSION

This sensitivity to higher frequencies make it a particularly useful tool to identify patients with the greatest disease burden. This could be used in a number of settings. It could be used in clinical care to flag patients with a seizure burden above a typical threshold for the clinician’s attention. It could be used in cohort selection to identify patients to be enrolled in clinical trials for a new medication or to be recommended for surgery. It could be used to build a phenotypic profile of patients to be used in retrospective analyses to understand

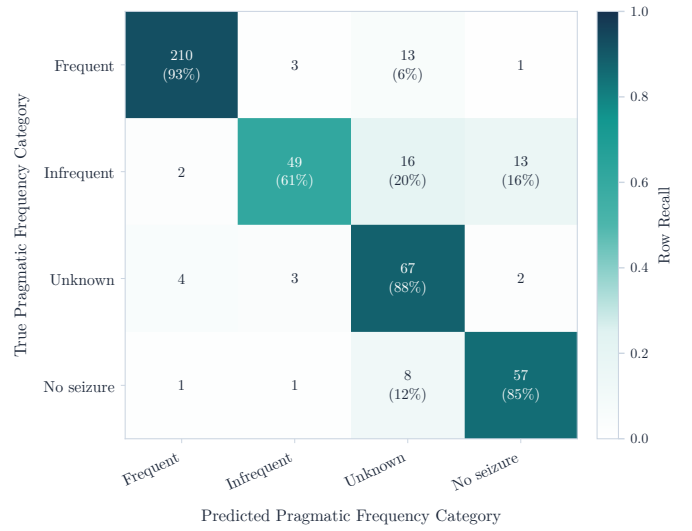


Fig. 6: The key error pattern is in conflating infrequent events with unknown or no seizure frequency references.

how these patients differ from the average. Finally, it could be used as an outcome variable in predictive models to identify patients likely to experience frequent seizures and recommend an early intervention.

VI. CONCLUSION

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Please number citations consecutively within brackets [1]. The sentence punctuation follows the bracket [2].

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